DG32-LITE - Preliminary Datasheet

Lockstep RISC-V motor-control SoC on SkyWater sky130A, in a 64-pin QFN. Dual rv32imc cores with a fault comparator, 32 KB SRAM, 64 KB self-boot ROM and QSPI-XIP, a 3-phase motor peripheral set (PWM, DShot, encoder, CORDIC, on-die SAR ADC) and UART/SPI/I2C/GPIO. For firmware and hardware/PCB engineers.

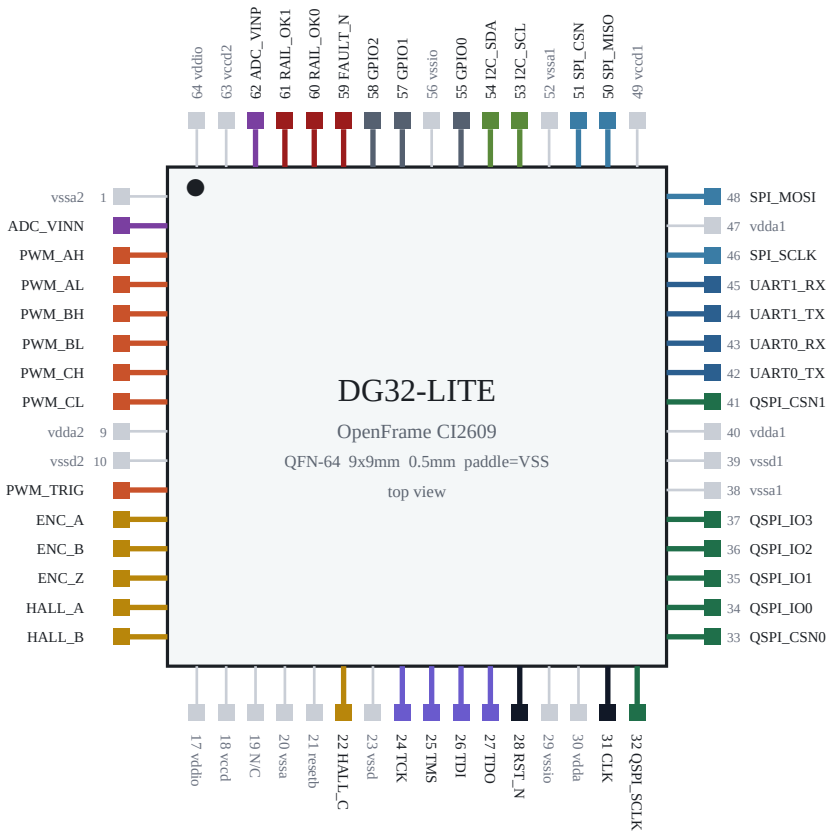
Part	Package	Process	Core	Clock
DG32-LITE	QFN-64, 9x9 mm	sky130A 130 nm	2x rv32imc lockstep	50 MHz

Preliminary, pre-silicon. Functional and register detail is from the taped-out RTL and the hardened die. Electrical limits are design targets and sky130 process nominals pending first-silicon characterisation, and are marked where they are not yet measured. The flight-control core is frozen until silicon test.

1 Features

- Dual DGridRiscV rv32imc cores in lockstep with a cycle-by-cycle fault comparator and a sticky fault CSR.
- 32 KB on-chip SRAM (no ECC); 64 KB baked boot ROM; QSPI execute-in-place window for external NOR flash.
- 3-phase center-aligned complementary PWM with programmable dead-time and a hardware brake.
- 4-channel DShot ESC engine, pin-muxed onto the PWM output pads.
- Quadrature + Hall encoder with edge timestamps; CORDIC (sin/cos, atan2, magnitude) for field-oriented control.
- On-die 8-bit differential SAR ADC for phase-current feedback, PWM-triggered.
- Two UARTs, an SPI master, a 7-bit I2C master, 14-pin GPIO, two timers, a windowed watchdog and rail supervisor.
- Interrupt controller; standalone self-boot (no management core); SLVERR on any unmapped access.
- Single 50 MHz clock domain; whole die powered from a 1.8 V rail; 3.3 V I/O.

2 Pinout - 64-pin QFN (top view)



Motor / PWM	Position sense	Analog ADC	QSPI flash
UART	SPI	I2C	GPIO
Supervisor	JTAG	Clock/Reset	Power / Gnd

Pin map derived from the OpenFrame harness general pinout and the DG32-LITE pad plan (rev 2), awaiting ChipFoundry bond-diagram confirmation. All GPIO pads are 3.3 V CMOS; the two analog pads (VINP, VINN) carry the SAR inputs and must stay within 0 to 1.8 V.

3 Pin description

QFN pin 1 is at the upper-left; numbering runs counter-clockwise. User signals are the DG32 function on each OpenFrame user pad; power and ground pins are the harness pad-ring supplies. The exposed paddle is VSS.

Pin	Signal	Dir	Note
1	**vssa2**	power	ground
2	**ADC_VINN**	analog	SAR ADC negative input, 0 to 1.8 V
3	**PWM_AH**	out	phase A high-side gate
4	**PWM_AL**	out	phase A low-side gate
5	**PWM_BH**	out	phase B high-side gate
6	**PWM_BL**	out	phase B low-side gate
7	**PWM_CH**	out	phase C high-side gate
8	**PWM_CL**	out	phase C low-side gate
9	**vdda2**	power	3.3 V user supply, unused by DG32-LITE — tie to 3.3 V
10	**vssd2**	power	ground
11	**PWM_TRIG**	out	ADC sample trigger (centre-aligned)
12	**ENC_A**	in	quadrature encoder A
13	**ENC_B**	in	quadrature encoder B
14	**ENC_Z**	in	encoder index
15	**HALL_A**	in	Hall sensor A
16	**HALL_B**	in	Hall sensor B
17	**vddio**	power	3.3 V pad-ring / ESD supply — REQUIRED
18	**vccd**	power	1.8 V harness core supply — REQUIRED
19	**N/C**	power	no connect
20	**vssa**	power	ground
21	**resetb**	power	harness reset input, active low, 3.3 V — ANDed into the DG32-LITE core reset; may be left pulled up if RST_N (pin 28) is used
22	**HALL_C**	in	Hall sensor C
23	**vssd**	power	ground
24	**TCK**	in (weak pull-down)	JTAG clock
25	**TMS**	in (weak pull-up)	JTAG mode select
26	**TDI**	in (weak pull-up)	JTAG data in
27	**TDO**	out (driven only while shifting)	JTAG data out
28	**RST_N**	in	external reset, active low (ANDed on die with resetb and the harness POR)
29	**vssio**	power	ground
30	**vdda**	power	3.3 V harness analog supply (POR) — REQUIRED
31	**CLK**	in	system clock, 3.3 V CMOS, 48 MHz nominal (core clock)
32	**QSPI_SCLK**	out	QSPI boot flash clock
33	**QSPI_CSN0**	out	QSPI flash chip select 0 (boot flash)
34	**QSPI_IO0**	bidir	QSPI data 0
35	**QSPI_IO1**	bidir	QSPI data 1
36	**QSPI_IO2**	bidir	QSPI data 2
37	**QSPI_IO3**	bidir	QSPI data 3
38	**vssa1**	power	ground
39	**vssd1**	power	ground (user area)
40	**vdda1**	power	3.3 V user supply, unused by DG32-LITE — tie to 3.3 V
41	**QSPI_CSN1**	out	QSPI chip select 1 (second device)
42	**UART0_TX**	out	console UART transmit; boot banner DG32" appears here"
43	**UART0_RX**	in	console UART receive
44	**UART1_TX**	out	UART 1 transmit
45	**UART1_RX**	in	UART 1 receive
46	**SPI_SCLK**	out	SPI master clock (external ADC / driver)
47	**vdda1**	power	3.3 V user supply, unused by DG32-LITE — tie to 3.3 V
48	**SPI_MOSI**	out	SPI master data out
49	**vccd1**	power	1.8 V user-area supply: all DG32-LITE logic, SRAM and the SAR ADC — REQUIRED
50	**SPI_MISO**	in	SPI master data in
51	**SPI_CSN**	out	SPI master chip select
52	**vssa1**	power	ground
53	**I2C_SCL**	bidir (open-drain)	I2C clock; external pull-up required
54	**I2C_SDA**	bidir (open-drain)	I2C data; external pull-up required
55	**GPIO0**	bidir	general-purpose I/O 0
56	**vssio**	power	ground
57	**GPIO1**	bidir	general-purpose I/O 1
58	**GPIO2**	bidir	general-purpose I/O 2
59	**FAULT_N**	out	lockstep / supervisor fault, active low
60	**RAIL_OK0**	in	rail supervisor input 0 (rail_ok[3:2] tied high on die)
61	**RAIL_OK1**	in	rail supervisor input 1
62	**ADC_VINP**	analog	SAR ADC positive input, 0 to 1.8 V (analog pad, digital buffer off)
63	**vccd2**	power	1.8 V user supply, unused by DG32-LITE — tie to 1.8 V
64	**vddio**	power	3.3 V pad-ring / ESD supply — REQUIRED

3.1 Pins by function

Function	QFN pins
Clock / reset	CLK=31, RST_N=28, resetb=21

Function	QFN pins
QSPI boot flash	SCLK=32, CSN0=33, IO0=34, IO1=35, IO2=36, IO3=37, CSN1=41
UART	UART0 TX=42 RX=43; UART1 TX=44 RX=45
SPI master	SCLK=46, MOSI=48, MISO=50, CSN=51
I2C	SCL=53, SDA=54 (open-drain, external pull-ups)
GPIO	GPIO0=55, GPIO1=57, GPIO2=58
Supervisor	FAULT_N=59, RAIL_OK0=60, RAIL_OK1=61
SAR ADC (analog)	VINP=62, VINN=2 (0 to 1.8 V)
Motor PWM	AH=3, AL=4, BH=5, BL=6, CH=7, CL=8, TRIG=11
Position sensing	ENC_A=12, ENC_B=13, ENC_Z=14, HALL_A=15, HALL_B=16, HALL_C=22
JTAG	TCK=24, TMS=25, TDI=26, TDO=27

4 Electrical characteristics

Pre-silicon. Recommended operating values are the sky130 process nominals for this die's supply domains; absolute-maximum values are conservative process limits, not device characterisation. First-silicon characterisation will replace this section.

4.1 Recommended operating conditions

Parameter	Sym	Min	Typ	Max	Unit	Note
Core supply	vccd1	1.71	1.8	1.89	V	sky130 core, +/-5%; powers all logic, SRAM and the SAR ADC
I/O supply	vddio	3.0	3.3	3.6	V	3.3 V CMOS pad ring / ESD
Core clock	CLK	-	50	~55	MHz	fmax ~55-62 MHz measured post-route; 50 MHz design target
ADC input	VINP/VINN	0	-	1.8	V	differential, vccd1 domain
Junction temp	TJ	-	25	-	degC	power quoted at tt 25 C; range TBD on characterisation

4.2 Absolute maximum ratings (process limits, pre-characterisation)

Parameter	Min	Max	Unit
Core supply vccd1	-0.3	1.95	V
I/O supply vddio	-0.3	3.63	V
Voltage on any digital pad	-0.3	vddio + 0.3	V
Analog pad voltage	-0.3	vccd1 + 0.3	V
Storage temperature	-40	125	degC (target)

4.3 Power

Estimated total ~0.43 W at 50 MHz (OpenROAD vectorless, tt 25 C, 1.8 V). All dynamic and leakage current is drawn from vccd1; the other user supply domains are unused by DG32-LITE and are tied only to keep the pad ring's domain-order ESD structures biased.

5 Power supplies and sequencing

Rail	QFN pins	Voltage	Purpose
vddio	17, 64	3.3 V	pad-ring / ESD supply - REQUIRED
vccd	18	1.8 V	harness core supply - REQUIRED
vccd1	49	1.8 V	user-area supply: all logic, SRAM, SAR ADC - REQUIRED
vdda	30	3.3 V	harness analog / POR supply - REQUIRED
vdda1	40, 47	3.3 V	user analog, unused - tie to 3.3 V
vdda2	9	3.3 V	user analog, unused - tie to 3.3 V
vccd2	63	1.8 V	user digital, unused - tie to 1.8 V
vss* / paddle	1,10,20,23,29,39,52,56 ...	0 V	all grounds and the exposed paddle

Sequencing. Bring up 3.3 V (pad ring) before or together with 1.8 V, as on the ChipFoundry development board. All vss* pins and the exposed paddle go to ground. DG32-LITE draws its whole current from vccd1 (1.8 V); tie the unused user rails (vdda1, vdda2, vccd2) to their nominal voltage so the ESD structures stay biased.

6 Clock, reset and boot

6.1 Clock and reset

A single core clock enters on CLK (pin 31), 3.3 V CMOS, 50 MHz design target. There is no on-die PLL. Two reset inputs are ANDed on die: the OpenFrame harness `resetb` (pin 21, active low) and `RST_N` (pin 28, active low); either may be used, pull the unused one high. The DG32-LITE die runs entirely from this one 50 MHz domain.

6.2 Boot flow

The die is standalone - no management core. At power-on the on-die 64 KB boot ROM runs with no external help:

- Emit the boot banner on UART0 (pin 42) at 115200 8N1 (baud divider fixed for 50 MHz).
- Read the QSPI NOR flash header through the 0x1 window (boot magic 0xD632_B007, then length).
- Copy the flash image into SRAM over the QSPI quad-output-fast-read (0x6B, 1-1-4, 8 dummy cycles, SCLK = clk/2).
- Jump to the copied image and run from SRAM/ROM.
- A UART monitor (r/w/j/i commands) is available if the flash is blank, so a bare board is still inspectable.

> **Instruction fetch is ROM-only.** The CPU fetch bus is wired to the boot ROM alone. The QSPI-XIP window and SRAM are reachable by loads and stores, not by instruction fetch - code does not execute in place from external NOR, it is copied to SRAM first. Firmware and linker scripts must place executable code in SRAM.

7 Memory map

Base-address decode from the top-level decoder: selector is addr[31:28]; the 0xD page sub-decodes addr[27:24], so each 0xD peripheral base is 0xDX00_0000. Any unmapped access completes with SLVERR (it never hangs the bus).

Base	Block	Note
0x1000_0000	QSPI-XIP read window	data read window (loads); NOT instruction fetch
0x9000_0000	SRAM	32 KB main data memory
0xC000_0000	DShot ESC engine	pin-muxed onto the PWM pads
0xD000_0000	UART0	console / boot banner
0xD100_0000	UART1	telemetry
0xD200_0000	SPI master	
0xD300_0000	GPIO	14 pins on the die
0xD400_0000	TIMER0	
0xD500_0000	TIMER1	
0xD600_0000	Fault CSR	poison + cause
0xD700_0000	Supervisor / wWDT	
0xD800_0000	PWM	3-phase center-aligned
0xD900_0000	I2C master	
0xDA00_0000	Encoder	quadrature + Hall
0xDB00_0000	SAR ADC	8-bit differential
0xDC00_0000	DMA	
0xDD00_0000	SYSCTL	device ID / clock gate
0xDE00_0000	CORDIC	Q1.31
0xDF00_0000	IRQ aggregator	

Address constraint (firmware). The CPU has a hardwired PMA whitelist: a load or store is legal only if addr[31] is set or the address is in 0x1xxx_xxxx. That is why SRAM is at 0x9 and the peripherals at 0xC/0xD. An access outside the whitelist faults inside the CPU before the bus sees it.

8 Register map

Byte offsets within each block, extracted from the RTL and cross-checked against the AXI decode. Sticky bits clear as noted (W1C = write 1 to clear). Self-clearing start bits are marked in the block.

UART0 / UART1 (0xD0 / 0xD1)

Offset	Name	R/W	Bits / meaning
0x00	**CTRL**	R/W	[0]=enable; [31:16]=clkdiv (baud = clk_i/clkdiv)
0x04	**TX**	W	[7:0]=byte to send (8N1); write back-pressures while tx_busy - never drops a byte
0x08	**RX**	R	[7:0]=received byte; read clears rx_valid
0x0C	**STATUS**	R	[0]=tx_busy; [1]=rx_valid

8N1: 1 start + 8 data LSB-first + 1 stop. RX 2-flop synchronised. 115200 baud at 50 MHz = 434 clocks/bit.

SPI master (0xD2)

Offset	Name	R/W	Bits / meaning
0x00	**CTRL**	R/W	[0]=enable; [1]=cpol; [2]=cpha; [9:4]=xfer_bits (8/16/32); [31:16]=clkdiv
0x04	**TX**	W/R	write launches a full-duplex MSB-first transfer when enabled and not busy
0x08	**RX**	R	last received word
0x0C	**STATUS**	R	[0]=busy
0x10	**CS**	R/W	[0]=cs_n level (reset 1 = deasserted)

SCLK idle level = CPOL. Transfer = 2 x xfer_bits SCLK edges. Hardware master, 25 MHz, 0.64 us per 16-bit word.

GPIO (0xD3)

Offset	Name	R/W	Bits / meaning
0x00	**DATA_OUT**	R/W	[N-1:0] value driven on pins whose OE bit is set
0x04	**OE**	R/W	[N-1:0] 1=drive, 0=high-Z input (reset 0 = all inputs); active-high
0x08	**DATA_IN**	R	[N-1:0] 2-flop synchronised pin state; writes accepted and discarded
0x0C	**SET**	W1S/R	write 1 sets that DATA_OUT bit; reads as DATA_OUT
0x10	**CLR**	W1C/R	write 1 clears that DATA_OUT bit; reads as DATA_OUT

SET/CLR are separate addresses onto one DATA_OUT register (atomic single-pin toggling). N=14 on the die.

TIMER0 / TIMER1 (0xD4 / 0xD5)

Offset	Name	R/W	Bits / meaning
0x00	**CTRL**	R/W	[0]=enable; [1]=auto_reload; [2]=irq_enable; write [8]=clear pending (W1C)
0x04	**RELOAD**	R/W	period-1 (fires every RELOAD+1 clocks)
0x08	**COUNT**	R	current down-counter value
0x0C	**STATUS**	R	[0]=pending (cleared via CTRL[8])

Down-counter; pulses at 0; irq = pending and irq_enable. Enable-rising reloads count.

Fault CSR (0xD6)

Offset	Name	R/W	Bits / meaning
0x0	**STATUS**	R	[0]=poison; [3:1]=cause. Sticky; cleared only by reset
0x4	**INJECT**	W	write MAGIC 0xDEAD_5AFE -> corrupt pulses 32 cycles. Reads 0
0x8	**EVENTS**	R	matched lockstep-compare count
0xC	**ID**	R	0xD632_0001

cause 001 = DATA_MISMATCH. Sticky-first: holds the first cause across continued execution.

Supervisor / wWDT (0xD7)

Offset	Name	R/W	Bits / meaning
0x00	**ID**	R	0x00D6_0006
0x04	**CTRL**	R/W	[0]=global_en; [1]=wdt_en; [7:4]=rail_en[3:0]
0x08	**DEGLITCH**	R/W	[7:0] deglitch cycles (reset 8)
0x0C	**HOLDOFF**	R/W	[15:0] reset-release holdoff (reset 1000)
0x10	**WDTMIN**	R/W	[15:0] watchdog window min (reset 100)
0x14	**WDTMAX**	R/W	[15:0] watchdog window max (reset 50000)
0x18	**KICK**	W	write 0xA5A5 = valid watchdog kick
0x1C	**STATUS**	R/W1C	[0]=sys_rst_n; [1]=wdt_fault; [7:4]=rail_sticky; [15:8]=fault_cnt

Windowed watchdog: a kick before WDTMIN or after WDTMAX both fault. Arms only when firmware sets global_en.

PWM (3-phase) (0xD8)

Offset	Name	R/W	Bits / meaning
0x00	**CTRL**	R/W	[0]=EN
0x04	**PERIOD**	R/W	counter top, 0..65535 (period = 2*(PERIOD+1) cycles, center-aligned)
0x08	**DEADTIME**	R/W	dead-time cycles, all 3 legs (8-bit, max 255 = 5.1 us at 50 MHz)
0x0C	**DUTY_A**	R/W	phase A shadow duty, 0..PERIOD
0x10	**DUTY_B**	R/W	phase B shadow duty
0x14	**DUTY_C**	R/W	phase C shadow duty
0x18	**STATUS**	R	[0]=BRAKED (sticky)
0x1C	**BRAKE_CLR**	W	write [0]=1 to clear BRAKED once brake_i is low again

Shadow duties latch at cnt==0; sample_trig pulses there. Async hardware brake forces all 6 outputs off in <=2 cycles.

I2C master (0xD9)

Offset	Name	R/W	Bits / meaning
0x00	**CTRL**	R/W	[0]=enable; [31:16]=clkdiv (SCL = clk_i/(4*(clkdiv+1)); reset 124 -> 100 kHz)
0x04	**CMD**	W/R	[7:1]=7-bit addr; [0]=read/write; [15:8]=byte count 1..4; [16]=send STOP
0x08	**TXD**	R/W	up to 4 bytes to send, LSB first on the wire
0x0C	**RXD**	R	up to 4 bytes received, LSB first
0x10	**STATUS**	R/W1C	[0]=busy; [1]=nack_addr; [2]=nack_data; [3]=arb_lost (write 1 to clear)

7-bit, 100/400 kHz, clock stretching, open-drain (external pull-ups). 2-byte read ~72 us at 400 kHz.

Encoder (quad + Hall) (0xDA)

Offset	Name	R/W	Bits / meaning
0x00	**CTRL**	R/W	[0]=EN (gates quad decode + Hall capture)
0x04	**POSITION**	R/W	signed 4x quadrature position; firmware may write to preset/zero
0x08	**INDEX_POSITION**	R	POSITION latched at last Z rising edge
0x0C	**STATUS**	R/W1C	[0]=INDEX_SEEN (sticky); [1]=DIR (live)
0x10	**TIMESTAMP**	R	free-running 32-bit cycle counter
0x14	**HALL_STATE**	R	[2:0]=synchronised {hall_c,hall_b,hall_a}
0x18	**HALL_TIMESTAMP**	R	TIMESTAMP at last Hall transition
0x1C	**HALL_EVENT**	R/W1C	[0]=NEW_EVENT (sticky)

Edge timestamps give velocity from time-between-edges, accurate at low speed.

SAR ADC (0xDB)

Offset	Name	R/W	Bits / meaning
0x00	**CTRL**	R/W	[0]=enable; [1]=cal; [31:16]=clkdiv (reset 24 -> 50 MHz/25 = 2 MHz)
0x04	**START**	W	any write starts one conversion when enabled and not busy
0x08	**RESULT**	R	[7:0]=last conversion; [8]=valid (cleared on START)
0x0C	**STATUS**	R	[0]=busy; [1]=valid; [2]=comp (raw comparator)
0x10	**TRIM**	R/W	[4:0]=trim bits (reset 01111). NOTE: no effect on silicon - see block note

8-bit differential SAR hard macro (1.8 V, on vccd1). ~200 kSa/s at the reset divider. The macro's trim/clkc/comp taps are walled inside its own metal and are left unrouted (routing them makes the die unroutable), so TRIM and CLKC have no effect on this silicon and STATUS.comp reads 0 - the ADC runs untrimmed. The register fields are kept for API stability.

DShot ESC engine (0xC000_0000)

Offset	Name	R/W	Bits / meaning
0x00	**CTRL**	W	[0]=START (self-clearing; ignored while BUSY)
0x04	**BITPERIOD**	R/W	cycles per bit slot (12-bit)
0x08	**T0H**	R/W	high-time for logical 0
0x0C	**T1H**	R/W	high-time for logical 1
0x10	**THROTTLE0**	R/W	[10:0]=throttle; [11]=telemetry request; channel 0
0x14	**THROTTLE1**	R/W	channel 1
0x18	**THROTTLE2**	R/W	channel 2
0x1C	**THROTTLE3**	R/W	channel 3
0x20	**STATUS**	R	[0]=BUSY

16-bit frame MSB-first: [15:5] throttle, [4] telemetry, [3:0] CRC4. All 4 channels share one bit-time counter.

CORDIC (0xDE)

Offset	Name	R/W	Bits / meaning
0x00	**CTRL**	R/W	[0]=START (self-clearing); [1]=MODE (0=rotation, 1=vectoring)
0x04	**XIN**	R/W	Q1.31 x input
0x08	**YIN**	R/W	Q1.31 y input
0x0C	**ZIN**	R/W	Q1.31 angle/pi
0x10	**XOUT**	R	Q1.31 result x (rotation: cos; vectoring: magnitude/K)
0x14	**YOUT**	R	Q1.31 result y (rotation: sin)
0x18	**ZOUT**	R	Q1.31 result z (vectoring: atan2/pi)
0x1C	**STATUS**	R	[0]=BUSY

N=20 iterations, ~53-58 cycles/op. K=0x4DBA76D4; vectoring magnitude is not gain-compensated by design.

IRQ aggregator (0xDF)

Offset	Name	R/W	Bits / meaning
0x00	**PENDING**	R/W1C	[15:0] pending; write 1 to clear
0x04	**ENABLE**	R/W	[15:0] mask; external int = OR(PENDING and ENABLE)
0x08	**MSIP**	R/W	[0] -> machine software interrupt
0x0C	**RAW**	R	[15:0] source levels before edge detection
0x10	**SWSET**	W	write 1 sets PENDING[7]
0x14	**ID**	R	0x00D6_1249

Sources: 0 TMR0, 1 TMR1, 2 PWM trig, 3 ADC done, 4 DMA done, 5 I2C done, 6 FAULT, 7 SW, 8 ATTN (2DOM).

DMA (0xDC)

Offset	Name	R/W	Bits / meaning
0x00	**CTRL**	R/W	[0]=enable; [1]=start (self-clearing); [2]=src_incr; [3]=dst_incr
0x04	**SRC**	R/W	source address, word-aligned
0x08	**DST**	R/W	destination address, word-aligned
0x0C	**LEN**	R/W	transfer length in words (0 = none)
0x10	**STATUS**	R/W1C	[0]=busy; [1]=done (write 1 to clear); [2]=error
0x14	**COUNT**	R	words remaining

Single-beat AXI master; the top arbitrates LSU-priority over DMA. ~7.1 cycles/word.

9 Board design guidance (hardware / PCB)

- Motor half-bridge gate driver on PWM_AH..CL (pins 3-8); the outputs are complementary with programmable dead-time. Respect the driver's shoot-through spec; the die also has a hardware brake that forces all six low.
- Route PWM_TRIG (pin 11) to nothing external unless you want the ADC-sample instant on a test point; it fires at the centre-aligned current-ripple null.
- Wire the encoder A/B/Z (pins 12-14) and Hall A/B/C (pins 15,16,22) directly; inputs are 2-flop synchronised and 3.3 V.
- SAR ADC inputs ADC_VINP (62) and ADC_VINN (2) are differential, 0 to 1.8 V - scale phase-current shunt amplifiers into that window and keep the pair matched and short.
- QSPI NOR boot flash on pins 32-37 (add CSN1=41 for a second device / PSRAM); a 3.3 V W25Q / S25FL works. Without a flash the ROM drops to the UART monitor.
- I2C_SCL (53) and I2C_SDA (54) are open-drain - the carrier must supply the pull-ups (typ 2.2-4.7 k to 3.3 V).
- Bring UART0 (42/43) to a USB-UART for the boot banner and console; UART1 (44/45) is free for telemetry.
- FAULT_N (59) is an active-low fault output from the lockstep/supervisor - route it to the driver enable or a supervisor for a hardware trip.
- Provide 3.3 V and 1.8 V per Section 5; the 100 production parts need a carrier with the 64-pin QFN footprint (KiCad from chipfoundry/caravel_board), gate drives, encoder, an optional external SPI ADC, and UART/JTAG headers.

JTAG idles safely with no probe. TMS (pin 25) and TDI (pin 26) are weakly pulled up on die, TCK (pin 24) weakly pulled down, so an unconnected JTAG header will not disturb the part.

10 Package

Item	Value
Package	64-pin QFN
Body	9.0 x 9.0 mm
Pitch	0.5 mm
Exposed paddle	VSS (solder to ground plane)
Pin 1	upper-left, counter-clockwise numbering
Process	SkyWater sky130A, 130 nm
Shuttle	chipIgnite CI2609

Preliminary datasheet, pre-silicon. Register and pin detail from the taped-out RTL (CI2609 R2) and the QFN-64 pin map derived from the OpenFrame harness pinout, awaiting ChipFoundry bond-diagram confirmation. Electrical limits are design/process nominals pending first-silicon characterisation.

DG32-2DOM - Preliminary Datasheet

The DG32-LITE motor-control SoC plus an INT8 attention accelerator on a second clock. Same lockstep core, boot path and motor peripheral set; adds a memory-mapped attention engine (0xE000_0000) that runs on clk_fast_i and reaches memory through clock-domain-crossing bridges. For firmware and hardware/PCB engineers.

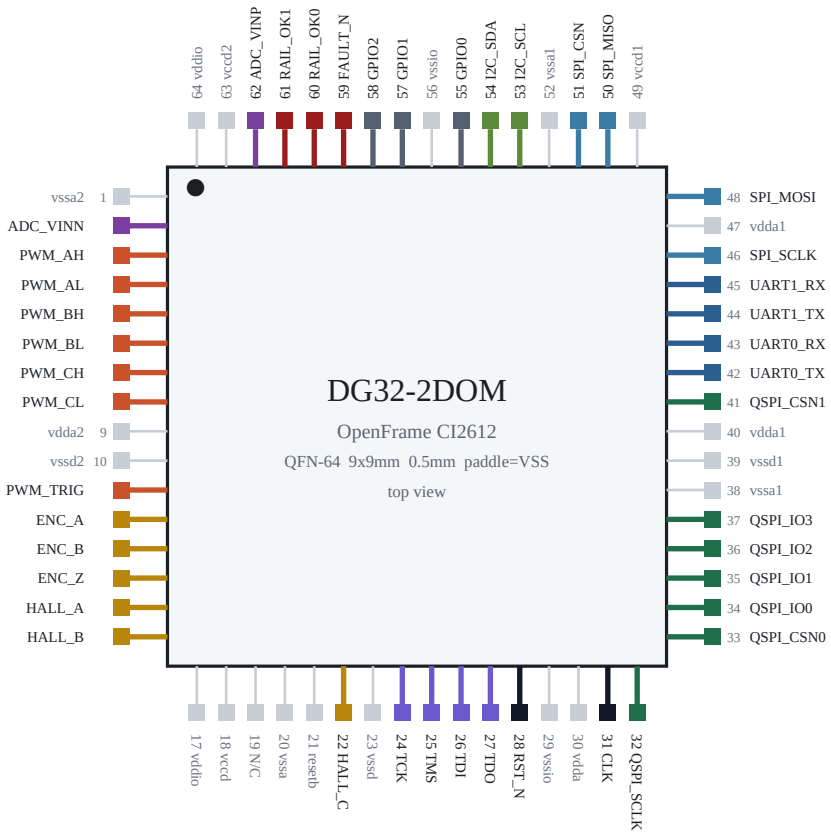
Part	Package	Process	Core	Clocks
DG32-2DOM	QFN-64, 9x9 mm	sky130A 130 nm	2x rv32imc + INT8 attn	50 / 114 MHz

Preliminary, pre-silicon. Functional and register detail is from the taped-out RTL and the hardened die. Electrical limits are design targets and sky130 process nominals pending first-silicon characterisation, and are marked where they are not yet measured. The flight-control core is frozen until silicon test.

1 Features

- Everything in DG32-LITE: dual rv32imc lockstep core, 32 KB SRAM, 64 KB boot ROM, QSPI-XIP, and the full motor peripheral set (PWM, DShot, encoder, CORDIC, SAR ADC) plus UART/SPI/I2C/GPIO.
- INT8 attention accelerator (0xE000_0000): one kick computes a band of query rows of one head end to end (QK^T, softmax/EXP, E.V, requantise to int8), bit-exact to the golden model.
- Attention datapath: shared LANES-wide MAC array, 256-entry u15 EXP table, 40-bit numerator (exact to NK=131072), one reciprocal per row; ~3242 cycles/row analytic at LANES=16.
- Second clock domain: the engine runs on clk_fast_i (114 MHz) and crosses to the 50 MHz CPU domain through 2-flop CDC bridges, so it never stalls the frozen control core.
- Own die (3400 x 4500 um) with the attention K/V buffers on sky130 SRAM macros next to the SAR analog corner.

2 Pinout - 64-pin QFN (top view)



Motor / PWM	Position sense	Analog ADC	QSPI flash
UART	SPI	I2C	GPIO
Supervisor	JTAG	Clock/Reset	Power / Gnd

The external pinout is the same 64-pin QFN as DG32-LITE: the attention engine is internal (it talks to memory over the on-chip bus and to the CPU over AXI-lite), so it adds no user pads. The only added clock, clk_fast_i, is a review item - either derived on-die from CLK or brought in on a spare pad for the CI2612 die.

3 Pin description

QFN pin 1 is at the upper-left; numbering runs counter-clockwise. User signals are the DG32 function on each OpenFrame user pad; power and ground pins are the harness pad-ring supplies. The exposed paddle is VSS.

Pin	Signal	Dir	Note
1	**vssa2**	power	ground
2	**ADC_VINN**	analog	SAR ADC negative input, 0 to 1.8 V
3	**PWM_AH**	out	phase A high-side gate
4	**PWM_AL**	out	phase A low-side gate
5	**PWM_BH**	out	phase B high-side gate
6	**PWM_BL**	out	phase B low-side gate
7	**PWM_CH**	out	phase C high-side gate
8	**PWM_CL**	out	phase C low-side gate
9	**vdda2**	power	3.3 V user supply, unused by DG32-LITE — tie to 3.3 V
10	**vssd2**	power	ground
11	**PWM_TRIG**	out	ADC sample trigger (centre-aligned)
12	**ENC_A**	in	quadrature encoder A
13	**ENC_B**	in	quadrature encoder B
14	**ENC_Z**	in	encoder index
15	**HALL_A**	in	Hall sensor A
16	**HALL_B**	in	Hall sensor B
17	**vddio**	power	3.3 V pad-ring / ESD supply — REQUIRED
18	**vccd**	power	1.8 V harness core supply — REQUIRED
19	**N/C**	power	no connect
20	**vssa**	power	ground
21	**resetb**	power	harness reset input, active low, 3.3 V — ANDed into the DG32-LITE core reset; may be left pulled up if RST_N (pin 28) is used
22	**HALL_C**	in	Hall sensor C
23	**vssd**	power	ground
24	**TCK**	in (weak pull-down)	JTAG clock
25	**TMS**	in (weak pull-up)	JTAG mode select
26	**TDI**	in (weak pull-up)	JTAG data in
27	**TDO**	out (driven only while shifting)	JTAG data out
28	**RST_N**	in	external reset, active low (ANDed on die with resetb and the harness POR)
29	**vssio**	power	ground
30	**vdda**	power	3.3 V harness analog supply (POR) — REQUIRED
31	**CLK**	in	system clock, 3.3 V CMOS, 48 MHz nominal (core clock)
32	**QSPI_SCLK**	out	QSPI boot flash clock
33	**QSPI_CSN0**	out	QSPI flash chip select 0 (boot flash)
34	**QSPI_IO0**	bidir	QSPI data 0
35	**QSPI_IO1**	bidir	QSPI data 1
36	**QSPI_IO2**	bidir	QSPI data 2
37	**QSPI_IO3**	bidir	QSPI data 3
38	**vssa1**	power	ground
39	**vssd1**	power	ground (user area)
40	**vdda1**	power	3.3 V user supply, unused by DG32-LITE — tie to 3.3 V
41	**QSPI_CSN1**	out	QSPI chip select 1 (second device)
42	**UART0_TX**	out	console UART transmit; boot banner DG32" appears here"
43	**UART0_RX**	in	console UART receive
44	**UART1_TX**	out	UART 1 transmit
45	**UART1_RX**	in	UART 1 receive
46	**SPI_SCLK**	out	SPI master clock (external ADC / driver)
47	**vdda1**	power	3.3 V user supply, unused by DG32-LITE — tie to 3.3 V
48	**SPI_MOSI**	out	SPI master data out
49	**vccd1**	power	1.8 V user-area supply: all DG32-LITE logic, SRAM and the SAR ADC — REQUIRED
50	**SPI_MISO**	in	SPI master data in
51	**SPI_CSN**	out	SPI master chip select
52	**vssa1**	power	ground
53	**I2C_SCL**	bidir (open-drain)	I2C clock; external pull-up required
54	**I2C_SDA**	bidir (open-drain)	I2C data; external pull-up required
55	**GPIO0**	bidir	general-purpose I/O 0
56	**vssio**	power	ground
57	**GPIO1**	bidir	general-purpose I/O 1
58	**GPIO2**	bidir	general-purpose I/O 2
59	**FAULT_N**	out	lockstep / supervisor fault, active low
60	**RAIL_OK0**	in	rail supervisor input 0 (rail_ok[3:2] tied high on die)
61	**RAIL_OK1**	in	rail supervisor input 1
62	**ADC_VINP**	analog	SAR ADC positive input, 0 to 1.8 V (analog pad, digital buffer off)
63	**vccd2**	power	1.8 V user supply, unused by DG32-LITE — tie to 1.8 V
64	**vddio**	power	3.3 V pad-ring / ESD supply — REQUIRED

3.1 Pins by function

Function	QFN pins
Clock / reset	CLK=31, RST_N=28, resetb=21

Function	QFN pins
QSPI boot flash	SCLK=32, CSN0=33, IO0=34, IO1=35, IO2=36, IO3=37, CSN1=41
UART	UART0 TX=42 RX=43; UART1 TX=44 RX=45
SPI master	SCLK=46, MOSI=48, MISO=50, CSN=51
I2C	SCL=53, SDA=54 (open-drain, external pull-ups)
GPIO	GPIO0=55, GPIO1=57, GPIO2=58
Supervisor	FAULT_N=59, RAIL_OK0=60, RAIL_OK1=61
SAR ADC (analog)	VINP=62, VINN=2 (0 to 1.8 V)
Motor PWM	AH=3, AL=4, BH=5, BL=6, CH=7, CL=8, TRIG=11
Position sensing	ENC_A=12, ENC_B=13, ENC_Z=14, HALL_A=15, HALL_B=16, HALL_C=22
JTAG	TCK=24, TMS=25, TDI=26, TDO=27
Attention (internal)	memory-mapped at 0xE000_0000; clk_fast_i is a review item

4 Electrical characteristics

Pre-silicon. Recommended operating values are the sky130 process nominals for this die's supply domains; absolute-maximum values are conservative process limits, not device characterisation. First-silicon characterisation will replace this section.

4.1 Recommended operating conditions

Parameter	Sym	Min	Typ	Max	Unit	Note
Core supply	vccd1	1.71	1.8	1.89	V	sky130 core, +/-5%; powers all logic, SRAM and the SAR ADC
I/O supply	vddio	3.0	3.3	3.6	V	3.3 V CMOS pad ring / ESD
Core clock	CLK	-	50	~55	MHz	fmax ~55-62 MHz measured post-route; 50 MHz design target
ADC input	VINP/VINN	0	-	1.8	V	differential, vccd1 domain
Junction temp	TJ	-	25	-	degC	power quoted at tt 25 C; range TBD on characterisation

4.2 Absolute maximum ratings (process limits, pre-characterisation)

Parameter	Min	Max	Unit
Core supply vccd1	-0.3	1.95	V
I/O supply vddio	-0.3	3.63	V
Voltage on any digital pad	-0.3	vddio + 0.3	V
Analog pad voltage	-0.3	vccd1 + 0.3	V
Storage temperature	-40	125	degC (target)

4.3 Power

Estimated total ~0.43 W at 50 MHz (OpenROAD vectorless, tt 25 C, 1.8 V). All dynamic and leakage current is drawn from vccd1; the other user supply domains are unused by DG32-2DOM and are tied only to keep the pad ring's domain-order ESD structures biased.

5 Power supplies and sequencing

Rail	QFN pins	Voltage	Purpose
vddio	17, 64	3.3 V	pad-ring / ESD supply - REQUIRED
vccd	18	1.8 V	harness core supply - REQUIRED
vccd1	49	1.8 V	user-area supply: all logic, SRAM, SAR ADC - REQUIRED
vdda	30	3.3 V	harness analog / POR supply - REQUIRED
vdda1	40, 47	3.3 V	user analog, unused - tie to 3.3 V
vdda2	9	3.3 V	user analog, unused - tie to 3.3 V
vccd2	63	1.8 V	user digital, unused - tie to 1.8 V
vss* / paddle	1,10,20,23,29,39,52,56 ...	0 V	all grounds and the exposed paddle

Sequencing. Bring up 3.3 V (pad ring) before or together with 1.8 V, as on the ChipFoundry development board. All vss* pins and the exposed paddle go to ground. DG32-2DOM draws its whole current from vccd1 (1.8 V); tie the unused user rails (vdda1, vdda2, vccd2) to their nominal voltage so the ESD structures stay biased.

6 Clock, reset and boot

6.1 Clock and reset

A single core clock enters on CLK (pin 31), 3.3 V CMOS, 50 MHz design target. There is no on-die PLL. Two reset inputs are ANDed on die: the OpenFrame harness `resetb` (pin 21, active low) and `RST_N` (pin 28, active low); either may be used, pull the unused one high. DG32-2DOM adds a second clock, clk_fast_i (114 MHz), for the attention engine; the CPU and all peripherals stay in the 50 MHz domain and the two are bridged by CDC logic.

6.2 Boot flow

The die is standalone - no management core. At power-on the on-die 64 KB boot ROM runs with no external help:

- Emit the boot banner on UART0 (pin 42) at 115200 8N1 (baud divider fixed for 50 MHz).
- Read the QSPI NOR flash header through the 0x1 window (boot magic 0xD632_B007, then length).
- Copy the flash image into SRAM over the QSPI quad-output-fast-read (0x6B, 1-1-4, 8 dummy cycles, SCLK = clk/2).
- Jump to the copied image and run from SRAM/ROM.
- A UART monitor (r/w/j/i commands) is available if the flash is blank, so a bare board is still inspectable.

Instruction fetch is ROM-only. The CPU fetch bus is wired to the boot ROM alone. The QSPI-XIP window and SRAM are reachable by loads and stores, not by instruction fetch - code does not execute in place from external NOR, it is copied to SRAM first. Firmware and linker scripts must place executable code in SRAM.

7 Memory map

Base-address decode from the top-level decoder: selector is addr[31:28]; the 0xD page sub-decodes addr[27:24], so each 0xD peripheral base is 0xDX00_0000. Any unmapped access completes with SLVERR (it never hangs the bus).

Base	Block	Note
0x1000_0000	QSPI-XIP read window	data read window (loads); NOT instruction fetch
0x9000_0000	SRAM	32 KB main data memory
0xC000_0000	DShot ESC engine	pin-muxed onto the PWM pads
0xD000_0000	UART0	console / boot banner
0xD100_0000	UART1	telemetry
0xD200_0000	SPI master	
0xD300_0000	GPIO	14 pins on the die
0xD400_0000	TIMER0	
0xD500_0000	TIMER1	
0xD600_0000	Fault CSR	poison + cause
0xD700_0000	Supervisor / wWDT	
0xD800_0000	PWM	3-phase center-aligned
0xD900_0000	I2C master	
0xDA00_0000	Encoder	quadrature + Hall
0xDB00_0000	SAR ADC	8-bit differential
0xDC00_0000	DMA	
0xDD00_0000	SYSCTL	device ID / clock gate
0xDE00_0000	CORDIC	Q1.31
0xDF00_0000	IRQ aggregator	
0xE000_0000	INT8 attention	DG32-2DOM only (INCLUDE_ATTN)

Address constraint (firmware). The CPU has a hardwired PMA whitelist: a load or store is legal only if addr[31] is set or the address is in 0x1xxx_xxxx. That is why SRAM is at 0x9 and the peripherals at 0xC/0xD/0xE. An access outside the whitelist faults inside the CPU before the bus sees it.

8 Register map

Byte offsets within each block, extracted from the RTL and cross-checked against the AXI decode. Sticky bits clear as noted (W1C = write 1 to clear). Self-clearing start bits are marked in the block.

UART0 / UART1 (0xD0 / 0xD1)

Offset	Name	R/W	Bits / meaning
0x00	**CTRL**	R/W	[0]=enable; [31:16]=clkdiv (baud = clk_i/clkdiv)
0x04	**TX**	W	[7:0]=byte to send (8N1); write back-pressures while tx_busy - never drops a byte
0x08	**RX**	R	[7:0]=received byte; read clears rx_valid
0x0C	**STATUS**	R	[0]=tx_busy; [1]=rx_valid

8N1: 1 start + 8 data LSB-first + 1 stop. RX 2-flop synchronised. 115200 baud at 50 MHz = 434 clocks/bit.

SPI master (0xD2)

Offset	Name	R/W	Bits / meaning
0x00	**CTRL**	R/W	[0]=enable; [1]=cpol; [2]=cpha; [9:4]=xfer_bits (8/16/32); [31:16]=clkdiv
0x04	**TX**	W/R	write launches a full-duplex MSB-first transfer when enabled and not busy
0x08	**RX**	R	last received word
0x0C	**STATUS**	R	[0]=busy
0x10	**CS**	R/W	[0]=cs_n level (reset 1 = deasserted)

SCLK idle level = CPOL. Transfer = 2 x xfer_bits SCLK edges. Hardware master, 25 MHz, 0.64 us per 16-bit word.

GPIO (0xD3)

Offset	Name	R/W	Bits / meaning
0x00	**DATA_OUT**	R/W	[N-1:0] value driven on pins whose OE bit is set
0x04	**OE**	R/W	[N-1:0] 1=drive, 0=high-Z input (reset 0 = all inputs); active-high
0x08	**DATA_IN**	R	[N-1:0] 2-flop synchronised pin state; writes accepted and discarded
0x0C	**SET**	W1S/R	write 1 sets that DATA_OUT bit; reads as DATA_OUT
0x10	**CLR**	W1C/R	write 1 clears that DATA_OUT bit; reads as DATA_OUT

SET/CLR are separate addresses onto one DATA_OUT register (atomic single-pin toggling). N=14 on the die.

TIMER0 / TIMER1 (0xD4 / 0xD5)

Offset	Name	R/W	Bits / meaning
0x00	**CTRL**	R/W	[0]=enable; [1]=auto_reload; [2]=irq_enable; write [8]=clear pending (W1C)
0x04	**RELOAD**	R/W	period-1 (fires every RELOAD+1 clocks)
0x08	**COUNT**	R	current down-counter value
0x0C	**STATUS**	R	[0]=pending (cleared via CTRL[8])

Down-counter; pulses at 0; irq = pending and irq_enable. Enable-rising reloads count.

Fault CSR (0xD6)

Offset	Name	R/W	Bits / meaning
0x0	**STATUS**	R	[0]=poison; [3:1]=cause. Sticky; cleared only by reset
0x4	**INJECT**	W	write MAGIC 0xDEAD_5AFE -> corrupt pulses 32 cycles. Reads 0
0x8	**EVENTS**	R	matched lockstep-compare count
0xC	**ID**	R	0xD632_0001

cause 001 = DATA_MISMATCH. Sticky-first: holds the first cause across continued execution.

Supervisor / wWDT (0xD7)

Offset	Name	R/W	Bits / meaning
0x00	**ID**	R	0x00D6_0006
0x04	**CTRL**	R/W	[0]=global_en; [1]=wdt_en; [7:4]=rail_en[3:0]
0x08	**DEGLITCH**	R/W	[7:0] deglitch cycles (reset 8)
0x0C	**HOLDOFF**	R/W	[15:0] reset-release holdoff (reset 1000)
0x10	**WDTMIN**	R/W	[15:0] watchdog window min (reset 100)
0x14	**WDTMAX**	R/W	[15:0] watchdog window max (reset 50000)
0x18	**KICK**	W	write 0xA5A5 = valid watchdog kick
0x1C	**STATUS**	R/W1C	[0]=sys_rst_n; [1]=wdt_fault; [7:4]=rail_sticky; [15:8]=fault_cnt

Windowed watchdog: a kick before WDTMIN or after WDTMAX both fault. Arms only when firmware sets global_en.

PWM (3-phase) (0xD8)

Offset	Name	R/W	Bits / meaning
0x00	**CTRL**	R/W	[0]=EN
0x04	**PERIOD**	R/W	counter top, 0..65535 (period = 2*(PERIOD+1) cycles, center-aligned)
0x08	**DEADTIME**	R/W	dead-time cycles, all 3 legs (8-bit, max 255 = 5.1 us at 50 MHz)
0x0C	**DUTY_A**	R/W	phase A shadow duty, 0..PERIOD
0x10	**DUTY_B**	R/W	phase B shadow duty
0x14	**DUTY_C**	R/W	phase C shadow duty
0x18	**STATUS**	R	[0]=BRAKED (sticky)
0x1C	**BRAKE_CLR**	W	write [0]=1 to clear BRAKED once brake_i is low again

Shadow duties latch at cnt==0; sample_trig pulses there. Async hardware brake forces all 6 outputs off in <=2 cycles.

I2C master (0xD9)

Offset	Name	R/W	Bits / meaning
0x00	**CTRL**	R/W	[0]=enable; [31:16]=clkdiv (SCL = clk_i/(4*(clkdiv+1)); reset 124 -> 100 kHz)
0x04	**CMD**	W/R	[7:1]=7-bit addr; [0]=read/write; [15:8]=byte count 1..4; [16]=send STOP
0x08	**TXD**	R/W	up to 4 bytes to send, LSB first on the wire
0x0C	**RXD**	R	up to 4 bytes received, LSB first
0x10	**STATUS**	R/W1C	[0]=busy; [1]=nack_addr; [2]=nack_data; [3]=arb_lost (write 1 to clear)

7-bit, 100/400 kHz, clock stretching, open-drain (external pull-ups). 2-byte read ~72 us at 400 kHz.

Encoder (quad + Hall) (0xDA)

Offset	Name	R/W	Bits / meaning
0x00	**CTRL**	R/W	[0]=EN (gates quad decode + Hall capture)
0x04	**POSITION**	R/W	signed 4x quadrature position; firmware may write to preset/zero
0x08	**INDEX_POSITION**	R	POSITION latched at last Z rising edge
0x0C	**STATUS**	R/W1C	[0]=INDEX_SEEN (sticky); [1]=DIR (live)
0x10	**TIMESTAMP**	R	free-running 32-bit cycle counter
0x14	**HALL_STATE**	R	[2:0]=synchronised {hall_c,hall_b,hall_a}
0x18	**HALL_TIMESTAMP**	R	TIMESTAMP at last Hall transition
0x1C	**HALL_EVENT**	R/W1C	[0]=NEW_EVENT (sticky)

Edge timestamps give velocity from time-between-edges, accurate at low speed.

SAR ADC (0xDB)

Offset	Name	R/W	Bits / meaning
0x00	**CTRL**	R/W	[0]=enable; [1]=cal; [31:16]=clkdiv (reset 24 -> 50 MHz/25 = 2 MHz)
0x04	**START**	W	any write starts one conversion when enabled and not busy
0x08	**RESULT**	R	[7:0]=last conversion; [8]=valid (cleared on START)
0x0C	**STATUS**	R	[0]=busy; [1]=valid; [2]=comp (raw comparator)
0x10	**TRIM**	R/W	[4:0]=trim bits (reset 01111). NOTE: no effect on silicon - see block note

8-bit differential SAR hard macro (1.8 V, on vccd1). ~200 kSa/s at the reset divider. The macro's trim/clkc/comp taps are walled inside its own metal and are left unrouted (routing them makes the die unroutable), so TRIM and CLKC have no effect on this silicon and STATUS.comp reads 0 - the ADC runs untrimmed. The register fields are kept for API stability.

DShot ESC engine (0xC000_0000)

Offset	Name	R/W	Bits / meaning
0x00	**CTRL**	W	[0]=START (self-clearing; ignored while BUSY)
0x04	**BITPERIOD**	R/W	cycles per bit slot (12-bit)
0x08	**T0H**	R/W	high-time for logical 0
0x0C	**T1H**	R/W	high-time for logical 1
0x10	**THROTTLE0**	R/W	[10:0]=throttle; [11]=telemetry request; channel 0
0x14	**THROTTLE1**	R/W	channel 1
0x18	**THROTTLE2**	R/W	channel 2
0x1C	**THROTTLE3**	R/W	channel 3
0x20	**STATUS**	R	[0]=BUSY

16-bit frame MSB-first: [15:5] throttle, [4] telemetry, [3:0] CRC4. All 4 channels share one bit-time counter.

CORDIC (0xDE)

Offset	Name	R/W	Bits / meaning
0x00	**CTRL**	R/W	[0]=START (self-clearing); [1]=MODE (0=rotation, 1=vectoring)
0x04	**XIN**	R/W	Q1.31 x input
0x08	**YIN**	R/W	Q1.31 y input
0x0C	**ZIN**	R/W	Q1.31 angle/pi
0x10	**XOUT**	R	Q1.31 result x (rotation: cos; vectoring: magnitude/K)
0x14	**YOUT**	R	Q1.31 result y (rotation: sin)
0x18	**ZOUT**	R	Q1.31 result z (vectoring: atan2/pi)
0x1C	**STATUS**	R	[0]=BUSY

N=20 iterations, ~53-58 cycles/op. K=0x4DBA76D4; vectoring magnitude is not gain-compensated by design.

IRQ aggregator (0xDF)

Offset	Name	R/W	Bits / meaning
0x00	**PENDING**	R/W1C	[15:0] pending; write 1 to clear
0x04	**ENABLE**	R/W	[15:0] mask; external int = OR(PENDING and ENABLE)
0x08	**MSIP**	R/W	[0] -> machine software interrupt
0x0C	**RAW**	R	[15:0] source levels before edge detection
0x10	**SWSET**	W	write 1 sets PENDING[7]
0x14	**ID**	R	0x00D6_1249

Sources: 0 TMR0, 1 TMR1, 2 PWM trig, 3 ADC done, 4 DMA done, 5 I2C done, 6 FAULT, 7 SW, 8 ATTN (2DOM).

DMA (0xDC)

Offset	Name	R/W	Bits / meaning
0x00	**CTRL**	R/W	[0]=enable; [1]=start (self-clearing); [2]=src_incr; [3]=dst_incr
0x04	**SRC**	R/W	source address, word-aligned
0x08	**DST**	R/W	destination address, word-aligned
0x0C	**LEN**	R/W	transfer length in words (0 = none)
0x10	**STATUS**	R/W1C	[0]=busy; [1]=done (write 1 to clear); [2]=error
0x14	**COUNT**	R	words remaining

Single-beat AXI master; the top arbitrates LSU-priority over DMA. ~7.1 cycles/word.

INT8 attention engine (0xE000_0000) - programming (writes)

Offset	Name	R/W	Bits / meaning
0x00	**CTRL**	W	[0]=start (self-clearing); [1]=SKIP_LOAD (reuse EXP/K/V from previous kick)
0x04	**A_BASE**	W	Q base byte address
0x08	**A_PITCH**	W	Q position pitch (bytes)
0x0C	**B_BASE**	W	K base byte address
0x10	**B_PITCH**	W	K position pitch (bytes)
0x14	**D_BASE**	W	O (output) base byte address
0x18	**D_PITCH**	W	O position pitch (bytes)
0x1C	**M**	W	MROWS: query rows this kick (1..65535)
0x20	**N**	W	NK: keys (1..MAXK=400)
0x24	**K**	W	KD: Q/K head dim in bytes (mult of LANES and 4)
0x28	**SHIFT**	W	EXP index right-shift (6-bit)
0x2C	**C_BASE**	W	V base byte address
0x30	**C_PITCH**	W	V position pitch (bytes)
0x34	**DV**	W	output channels/head in bytes (mult of LANES and 4)
0x38	**E_BASE**	W	EXP table base (256 u15 entries)
0x3C	**RATIO**	W	u32 16.16 window constant; rq = floor((RATIO<<16)/Z)

INT8 attention engine (0xE000_0000) - status (reads)

Offset	Name	R/W	Bits / meaning
0x00	**STATUS**	R	[0]=busy; [1]=done; [2]=Z==0 seen; [3]=rq overflow; [4]=EXP clamp; [5]=geometry rejected
0x40	**Z**	R	Z of the last completed row
0x44	**rq**	R	rq of the last completed row

Programming a kick (firmware). Write the Q/K/V/O base and pitch, the geometry (M rows, N keys, KD, DV), SHIFT, the EXP table base and RATIO, then set CTRL[0]=start. Poll STATUS[0]=busy to 0; STATUS[1]=done. SKIP_LOAD (CTRL[1]) reuses the K/V/EXP buffers from the previous kick. The engine runs on clk_fast_i (114 MHz) and reaches memory through the CDC bridges, so it does not stall the 50 MHz CPU.

9 Board design guidance (hardware / PCB)

- Motor half-bridge gate driver on PWM_AH..CL (pins 3-8); the outputs are complementary with programmable dead-time. Respect the driver's shoot-through spec; the die also has a hardware brake that forces all six low.
- Route PWM_TRIG (pin 11) to nothing external unless you want the ADC-sample instant on a test point; it fires at the centre-aligned current-ripple null.
- Wire the encoder A/B/Z (pins 12-14) and Hall A/B/C (pins 15,16,22) directly; inputs are 2-flop synchronised and 3.3 V.
- SAR ADC inputs ADC_VINP (62) and ADC_VINN (2) are differential, 0 to 1.8 V - scale phase-current shunt amplifiers into that window and keep the pair matched and short.
- QSPI NOR boot flash on pins 32-37 (add CSN1=41 for a second device / PSRAM); a 3.3 V W25Q / S25FL works. Without a flash the ROM drops to the UART monitor.
- I2C_SCL (53) and I2C_SDA (54) are open-drain - the carrier must supply the pull-ups (typ 2.2-4.7 k to 3.3 V).
- Bring UART0 (42/43) to a USB-UART for the boot banner and console; UART1 (44/45) is free for telemetry.
- FAULT_N (59) is an active-low fault output from the lockstep/supervisor - route it to the driver enable or a supervisor for a hardware trip.
- Provide 3.3 V and 1.8 V per Section 5; the 100 production parts need a carrier with the 64-pin QFN footprint (KiCad from chipfoundry/caravel_board), gate drives, encoder, an optional external SPI ADC, and UART/JTAG headers.
- The attention engine adds no board signals; budget for clk_fast_i only if the CI2612 die brings it out to a pad rather than deriving it on-die.

JTAG idles safely with no probe. TMS (pin 25) and TDI (pin 26) are weakly pulled up on die, TCK (pin 24) weakly pulled down, so an unconnected JTAG header will not disturb the part.

10 Package

Item	Value
Package	64-pin QFN
Body	9.0 x 9.0 mm
Pitch	0.5 mm
Exposed paddle	VSS (solder to ground plane)
Pin 1	upper-left, counter-clockwise numbering
Process	SkyWater sky130A, 130 nm
Shuttle	chipIgnite CI2612

Preliminary datasheet, pre-silicon. Register and pin detail from the taped-out RTL (CI2609 R2) and the QFN-64 pin map derived from the OpenFrame harness pinout, awaiting ChipFoundry bond-diagram confirmation. Electrical limits are design/process nominals pending first-silicon characterisation. DG32-2DOM adds the INT8 attention engine and a second (114 MHz) clock domain; its A3 die closed with core +0.30 ns at 50 MHz and clk_fast +0.19 ns at 114 MHz.